



## Comprehensive Institutional Safety & Operational Platform

A specialized, hardware-accelerated School OS engineered for CBSE, ICSE, and SEBA private schools. Built to eliminate morning gate congestion, guarantee 100% cellular SMS delivery to parents, track school bus boarding, and automate faculty payroll.

1.2s

GATE KIOSK SCAN

14 Paise

CELLULAR SMS RATE

0% App

PARENT APP BARRIER

+₹2.28L

ANNUAL SCHOOL SURPLUS

### 01 Core Deliverables & Operational Modules

#### 1.2s High-Speed Gate Kiosk ( `/kiosk` )

Operates via industrial TVS/Honeywell 2D laser barcode scanners or webcams. Effortlessly scans **1,500+ students in 15 minutes** with zero morning congestion. Features dynamic encrypted QR codes to prevent proxy check-ins.

#### 100% Cellular Fast2SMS Gateway (14p/SMS)

Replaces fragile mobile app notifications with direct Transactional Cellular SMS delivered in **under 1 second** to any mobile phone (feature phone or smartphone) without requiring parents to install any app.

#### Bus Conductor Mobile Scanner ( `/bus` )

Lightweight mobile PWA for bus conductors. Features high-pitch audio and voice validation on boarding/drop-off, route passenger manifests, and an offline manual checklist fallback.

#### 2-3+ Days Consecutive Missing Inquest Radar

Automated safety tracking that detects students absent for 2, 3, or 5+ consecutive days. Features **1-click WhatsApp inquest notice dispatch**, reason tagging (Medical/Truancy), and official CBSE Rule 14 warning letters.

#### Faculty Attendance & Automated Payroll ( `/` )

Staff daily punch tracking with automated 1-click monthly payroll calculation: **EPF (12%)**, **Assam Professional Tax (₹208/mo)**, TDS, late deductions, and direct WhatsApp salary voucher dispatch.

#### ☐ Smart PVC ID Badge Studio & CBSE Register

Batch print CR80 PVC identity cards with emergency blood group and QR tokens ( `/badges` ). Export 1-click official monthly inspection registers compliant with CBSE By-Laws ( `/register` ).



02 Feature Matrix: Generic Cloud ERPs vs. SmartSchool OS

| EVALUATION PARAMETER               | GENERIC CLOUD ERPS<br>(SchoolAdmin.io, Teachmint, Entab, Fedena)                                                           | SMARTSCHOOL OS PLATFORM<br>(Our Ground Safety & Revenue Engine)                                                                 |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Gate Attendance Execution          | ❌ <b>Impractical at Gate:</b> Relies on teachers taking roll-call on app or students opening phones (which are banned).    | ✅ <b>1.2s Industrial Laser Kiosk:</b> Scans physical PVC QR badges at gate speed (100+ students/min). Zero congestion.          |
| Parent Alert Reliability           | ❌ <b>App Push Notifications:</b> Fails for 60%+ parents due to low phone storage, missing data packs, or uninstalled apps. | ✅ <b>100% Cellular SMS (14 Paise):</b> Sub-second SMS delivery directly to phone inbox on all cellular networks (Fast2SMS DLT). |
| School Bus Boarding Tracking       | ❌ <b>Not Supported</b> or requires expensive proprietary GPS hardware costing ₹15,000/bus.                                 | ✅ <b>Conductor Mobile App (‘/bus’):</b> 1-tap card scanning on any smartphone with audio feedback & offline fallback.           |
| Prolonged Absence Safety Radar     | ⚠️ <b>Basic Attendance Count:</b> Shows aggregate monthly % with no real-time streak detection.                            | ✅ <b>2-3+ Days Inquest Radar:</b> Flags safety risks, sends WhatsApp inquests, and prints CBSE Rule 14 letters.                 |
| Teacher Payroll & State Compliance | ⚠️ <b>Generic Payroll:</b> Requires manual entry; no regional Assam tax rules or payslip dispatch.                         | ✅ <b>Automated State Payroll:</b> 1-click salary slip with EPF (12%), Assam P-Tax (₹208), TDS & WhatsApp vouchers.              |
| Financial Model for School         | ❌ <b>Heavy Expense Item:</b> School pays ₹50,000 to ₹1,00,000+ subscription fee out of pocket every year.                  | ✅ <b>Surplus Profit Generator:</b> Generates + <b>₹2,28,801 net surplus annual profit</b> for the school.                       |
| Offline & Edge Resilience          | ❌ <b>Cloud-Only:</b> Completely freezes during local broadband or WiFi outages in tier-2/3 towns.                          | ✅ <b>Local-First Edge Architecture:</b> Scans and operates uninterrupted even if internet connection drops.                     |

03 The 3 Fatal Flaws of Generic ERPs at Indian School Gates

1. The Smartphone Dilemma

CBSE and State Education regulations **strictly ban student smartphones on campus**. Any attendance software requiring an app cannot operate for student check-in. Our durable PVC QR badges solve this completely.

2. The App Download Barrier

Less than 35% of Indian parents actively keep school apps installed due to low phone RAM, expired data packs, or lost passwords. Our **14-paise cellular SMS** guarantees 100% parent reach.

3. The 08:00 AM Gate Rush

1,500 students arrive within a narrow 30-minute window (07:45–08:15 AM). Biometric fingerprint scanners are too slow and unhygienic. Our **1.2s laser scanner** effortlessly clears 100+ scans per minute.



## 04 The Financial Transformation (1,500 Students Model)

### Annual School Revenue & Cost Ledger

|                                                         |            |
|---------------------------------------------------------|------------|
| Parent Digital Safety & Smart ID Fee (₹200/yr × 1,500): | +₹3,00,000 |
| Actual Annual Cellular SMS Cost (220 days × ₹0.14):     | -₹46,200   |
| Smart PVC ID Cards & System Configuration:              | -₹24,999   |

#### NET ANNUAL SURPLUS PROFIT FOR SCHOOL

**+₹2,28,801**

Direct Cash Surplus for School Development Fund

**Summary for Trustees:** Generic ERPs (e.g. SchoolAdmin.io) charge your school **-₹1,00,300/year** out of pocket. With SmartSchool OS, you charge a standard ₹200 annual Digital Safety Fee at admission, pay for all SMS and hardware, and earn **+₹2.28 Lakhs in pure profit** every year.

## 05 Turnkey 48-Hour Implementation Roadmap

### Day 1: Roster Upload

School provides Excel student/staff roster. System generates encrypted QR tokens and activates Fast2SMS DLT cellular header.

### Day 2: Badges & Scanner

CR80 Smart PVC Badges printed and distributed. Honeywell/TVS USB laser scanners installed at campus gates and bus routes.

### Day 3: Live Operations

1.2s gate check-in goes live. Parents begin receiving sub-second arrival SMS alerts and Principal War-Room activates.

## 06 Proposal Acceptance & Institutional Authorization

### Authorized Signatory / Principal

School Name & Seal: \_\_\_\_\_

### Technical Lead & Solutions Architect

SmartSchool OS Enterprise Infrastructure